

Kim Knudsen, Technical University of Denmark

Inverse Problems and Imaging (02624)

Week 3

Hilbert spaces and compact operators

Hilbert spaces X : Complete space with inner product: a bilinear map from $x, z \in X$ to $(x, z) \in \mathbb{R}$

Examples

- Euclidean space $X = \mathbb{R}^n$ with inner product $(x, z) = x \cdot z = x^T z$
- Function space $X = L^2(a, b)$ with inner product

$$(x, z) = \int_a^b x(s)z(s) \, ds.$$

Assume an ONB $\{x_j\}$ for X .

$$\text{Expansion: } x = \sum_j (x, x_j) x_j$$

$$\text{Norm: } \|x\|^2 = (x, x) = \sum_j |(x, x_j)|^2$$

Compact operator

$K \in B(X, Y)$ from Hilbert spaces X to Y is compact if it maps bounded sets to precompact sets.

Think of generalization of matrices to infinite dimensions.

Examples:

$$A \in \mathbb{R}^{n \times m}, \quad Kx(t) = \int_a^b k(t, s)x(s) \, ds \text{ in } L^2(a, b) \text{ with } \int \int |k(s, t)|^2 dt ds < \infty.$$

Singular Value Decomposition

X, Y : Hilbert spaces. Compact operator $K : X \rightarrow Y$.

Assume $\text{Dim}(X) = \infty$, $\text{Ker}(K) = \{0\}$.

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Singular Value Decomposition

$$\begin{aligned} \{x_j\} \text{ ONB for } X, \quad \{y_j\} \text{ ONB for } \overline{\text{Ran}(K)} \\ Kx_j = \mu_j y_j \end{aligned}$$

with

$$\mu_1 \geq \mu_2 \geq \cdots \geq \mu_j \geq \mu_{j+1} > 0, \text{ with } \lim_{j \rightarrow \infty} \mu_j = 0.$$

Expansion $x = \sum_j (x, x_j) x_j$ gives

$$Kx = \sum_j \mu_j (x, x_j) y_j.$$

Example: Numerical differentiation in $L^2(0, 1)$

$$Kx(t) = \int_0^t x(s) \, ds$$

Singular system

$$x_j(s) = \sqrt{2} \cos \left(\left(j - \frac{1}{2} \right) \pi s \right)$$

$$y_j(t) = \sqrt{2} \sin \left(\left(j - \frac{1}{2} \right) \pi t \right)$$

$$\mu_j = \frac{1}{\left(j - \frac{1}{2} \right) \pi}, \quad j = 1, 2, \dots$$

The role of noise

Data is always corrupted by noise

$$y^\delta = K\hat{x} + d = y + d$$

Absolute noise level $\|d\| = \delta$

Relative noise level $\frac{\|d\|}{\|y\|} = \frac{\delta}{\|y\|}$.

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Typically $d \notin \text{Ran}(K)$ since we may have

- 1 $\text{Proj}_{\text{Ker}K^*}(d) \neq 0$
- 2 $d - \text{Proj}_{\text{Ker}K^*}(d) \notin \text{Ran}(K)$

NB:

- Assume from now $\text{Ker}K^* = \{0\}$.
- In infinite dimensions, $\text{Ran}(K)$ is not closed.

Solving Inverse Problems by SVD

For $x = \sum_j (x, x_j) x_j$ consider $Kx = y$,

$$Kx = \sum_j \mu_j (x, x_j) y_j = \sum_j (y, y_j) y_j = y.$$

Solving

$$(y, y_j) = \mu_j (x, x_j) \quad (\text{for all } j); \quad x = \sum_j \frac{(y, y_j)}{\mu_j} x_j.$$

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Truncated SVD with $\alpha > 0$

$$x_\alpha = R_\alpha y = \sum_{\mu_j^2 \geq \alpha} \frac{(y, y_j)}{\mu_j} x_j.$$

With noise - the Picard condition

The equation $Kx = y^\delta = K\hat{x} + d$, i.e.

$$Kx = \sum_j (x, x_j) Kx_j = \sum_j \mu_j (x, x_j) y_j = \sum_j (y^\delta, y_j) y_j = y^\delta.$$

Solvable if and only if

$$y \in \text{Ran } K \Leftrightarrow Kx = y^\delta \Leftrightarrow \sum_j |(x, x_j)|^2 = \sum_j \frac{|(y^\delta, y_j)|^2}{\mu_j^2} < \infty.$$

Relation to the existence question?